
Θέματα Εξαμηνιαίας Εργασίας Αναγνώριση Προτύπων 2025-26

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Area 1: Interpretability

[INT-1] Emergent Introspective Awareness in Large Language Models

Link: <https://transformer-circuits.pub/2025/introspection/index.html>

We investigate whether large language models can introspect on their internal states. It is difficult to answer this question through conversation alone, as genuine introspection cannot be distinguished from confabulations. Here, we address this challenge by injecting representations of known concepts into a model's activations, and measuring the influence of these manipulations on the model's self-reported states. We find that models can, in certain scenarios, notice the presence of injected concepts and accurately identify them. Models demonstrate some ability to recall prior internal representations and distinguish them from raw text inputs. Strikingly, we find that some models can use their ability to recall prior intentions in order to distinguish their own outputs from artificial prefills. In all these experiments, Claude Opus 4 and 4.1, the most capable models we tested, generally demonstrate the greatest introspective awareness; however, trends across models are complex and sensitive to post-training strategies. Finally, we explore whether models can explicitly control their internal representations, finding that models can modulate their activations when instructed or incentivized to "think about" a concept. Overall, our results indicate that current language models possess some functional introspective awareness of their own internal states. We stress that in today's models, this capacity is highly unreliable and context-dependent; however, it may continue to develop with further improvements to model capabilities.

[INT-2] Concept Bottleneck Large Language Models

Link: <https://arxiv.org/pdf/2412.07992>

This paper introduces Concept Bottleneck Large Language Models (CB-LLMs), a framework for building inherently interpretable LLMs that provide transparency into their decision-making process. Unlike traditional black-box LLMs that rely on limited post-hoc interpretations, CB-LLMs integrate intrinsic interpretability directly into the model architecture by incorporating a human-interpretable Concept Bottleneck Layer (CBL) alongside a linear prediction layer.

[INT-3] Questioning Representational Optimism in Deep Learning: The Fractured Entangled Representation Hypothesis

Link: arxiv.org/pdf/2505.11581

The paper challenges the assumption that better performance necessarily implies better internal representations. It compares neural networks evolved through an open-ended search process to networks trained via conventional stochastic gradient descent on the simple task of generating a single image. While both networks produce the same output behavior, their internal representations differ dramatically. The SGD-trained networks exhibit a form of disorganization termed fractured entangled representation (FER). Interestingly, the evolved networks largely lack FER, even approaching a unified factored representation (UFR). In large models, FER may be degrading core model capacities like generalization, creativity, and continual learning.

[INT-4] When Models Manipulate Manifolds: The Geometry of a Counting Task

Link: <https://transformer-circuits.pub/2025/linebreaks/index.html>

Abstract: Intelligent systems need perception to understand, predict, and navigate their environment. These sensory capabilities reflect what's useful for survival in a specific environment: bats use echolocation, migratory birds sense magnetic fields, Arctic reindeer shift their UV vision seasonally. But when your world is made of text, what do you see? Language models encounter many text-based tasks that benefit from visual or spatial reasoning: parsing ASCII art, interpreting tables, or handling text wrapping constraints. Yet their only "sensory" input is a sequence of integers representing tokens. They must learn perceptual abilities from scratch, developing specialized mechanisms in the process. In this work, we investigate the mechanisms that enable Claude 3.5 Haiku to perform a natural perceptual task which is common in pretraining corpora and involves tracking position in a document. The task we study is linebreaking in fixed-width text. When training on source code, chat logs, email archives, scanned articles, or judicial rulings that have line width constraints, how does the model learn to predict when to break a line? We find learned representations of position that are in some ways quite similar to the biological neurons found in mammals who perform analogous tasks ("place cells" and "boundary cells" in mice), but in other ways unique to the constraints of the residual stream in language models.

[INT-5] Compression Represents Intelligence Linearly

Link: <https://arxiv.org/abs/2404.09937>

Abstract: There is a belief that learning to compress well will lead to intelligence. Recently, language modeling has been shown to be equivalent to compression, which offers a compelling rationale for the success of large language models (LLMs): the development of more advanced language models is essentially enhancing compression which facilitates intelligence. Despite such appealing discussions, little empirical evidence is present for the interplay between compression and intelligence. In this work, we examine their relationship in the context of LLMs, treating LLMs as data compressors. Given the abstract concept of "intelligence", we adopt the average downstream benchmark scores as a surrogate, specifically targeting intelligence related to

knowledge and commonsense, coding, and mathematical reasoning. Across 12 benchmarks, our study brings together 31 public LLMs that originate from diverse organizations. Remarkably, we find that LLMs' intelligence -- reflected by average benchmark scores -- almost linearly correlates with their ability to compress external text corpora. These results provide concrete evidence supporting the belief that superior compression indicates greater intelligence. Furthermore, our findings suggest that compression efficiency, as an unsupervised metric derived from raw text corpora, serves as a reliable evaluation measure that is linearly associated with the model capabilities. We open-source our compression datasets as well as our data collection pipelines to facilitate future researchers to assess compression properly.

Area 2: Large Language Models

[LLM-1] Gated Attention for Large Language Models

Link: <https://arxiv.org/abs/2505.06708>

Gating mechanisms have been widely utilized, from early models like LSTMs and Highway Networks to recent state space models, linear attention, and softmax attention. Yet, existing literature rarely examines the specific effects of gating. In this work, we conduct comprehensive experiments to systematically investigate gating-augmented softmax attention variants. Specifically, we perform a comprehensive comparison over 30 variants of 15B Mixture-of-Experts (MoE) models and 1.7B dense models trained on a 3.5 trillion token dataset. Our central finding is that a simple modification—applying a head-specific sigmoid gate after the Scaled Dot-Product Attention (SDPA)—consistently improves performance. This modification also enhances training stability, tolerates larger learning rates, and improves scaling properties. By comparing various gating positions and computational variants, we attribute this effectiveness to two key factors: (1) introducing non-linearity upon the low-rank mapping in the softmax attention, and (2) applying query-dependent sparse gating scores to modulate the SDPA output. Notably, we find this sparse gating mechanism mitigates 'attention sink' and enhances long-context extrapolation performance, and we also release related codes and models to facilitate future research.

[LLM-2] Let the Expert Stick to His Last: Expert-Specialized Fine-Tuning for Sparse Architectural Large Language Models

Link: <https://aclanthology.org/2024.emnlp-main.46.pdf>

Parameter-efficient fine-tuning (PEFT) is crucial for customizing Large Language Models (LLMs) with constrained resources. Although there have been various PEFT methods for dense-architecture LLMs, PEFT for sparse-architecture LLMs is still underexplored. In this work, we study the PEFT method for LLMs with the Mixture-of-Experts (MoE) architecture and the contents of this work are mainly threefold: (1) We investigate the dispersion degree of the activated experts in customized tasks, and found that the routing distribution for a specific task tends to be highly concentrated, while the distribution of activated experts varies significantly across different tasks. (2) We propose Expert-Specialized Fine-Tuning, or ESFT, which tunes the experts most relevant to downstream tasks while freezing the other experts and modules; experimental results

demonstrate that our method not only improves the tuning efficiency, but also matches or even surpasses the performance of full-parameter fine-tuning. (3) We further analyze the impact of the MoE architecture on expert-specialized fine-tuning. We find that MoE models with finer-grained experts are more advantageous in selecting the combination of experts that are most relevant to downstream tasks, thereby enhancing both the training efficiency and effectiveness.

[LLM-3] Learning Facts at Scale with Active Reading

Link: <https://arxiv.org/abs/2508.09494>

Abstract: LLMs are known to store vast amounts of knowledge in their parametric memory. However, learning and recalling facts from this memory is known to be unreliable, depending largely on the prevalence of particular facts in the training data and other factors which are poorly understood. Practitioners are lacking tools which will allow them to ensure that the models learn a given body of knowledge reliably and consistently. To this end, we propose Active Reading: a framework where we train models to study a given set of material with self-generated learning strategies. First, we demonstrate models trained with Active Reading on expert domains absorb significantly more knowledge than vanilla finetuning and other data augmentations. We train expert 8B models that achieve 66% on a Wikipedia-grounded subset of SimpleQA (+313% relative over vanilla finetuning) and 26% on FinanceBench (+160% relative over vanilla finetuning) by applying Active Reading to the source documents for each benchmark. Finally, we show that Active Reading can be utilized at pre-training scale to build more factual models. As a demonstration of this, we release Meta WikiExpert-8B, a Wikipedia-expert model trained on 1 trillion generated tokens, which outcompetes models with hundreds of billions of parameters on factual QA.

[LLM-4] LLM-AutoDiff: Auto-Differentiate Any LLM Workflow

Link: <https://arxiv.org/pdf/2501.16673v2>

Current prompt engineering methods for directing Large Language Models (LLMs) in complex workflows — like multi-step pipelines combining retrieval, reasoning, and formatting — require extensive manual tuning. This becomes especially cumbersome when tasks involve cyclic or repeated LLM calls (e.g., iterative refinement loops), leading to inefficiency and suboptimal performance in applications such as multi-hop question answering or autonomous agents. The authors developed LLM-AutoDiff, an automatic prompt engineering framework that treats textual prompts as trainable parameters. Inspired by neural network optimization, the framework uses a frozen "backward engine" LLM to generate textual feedback (akin to gradients) that iteratively refines prompts across multi-component workflows. This approach uniquely handles cyclic dependencies, isolates sub-prompts (e.g., instructions, formats) to prevent context dilution ("lost-in-the-middle"), and prioritizes error-prone samples for faster training. LLM-AutoDiff outperformed existing textual gradient methods in accuracy and efficiency across diverse tasks, including single-step classification and multi-hop retrieval workflows. Notably, it improved performance in agent-driven pipelines and resolved bottlenecks in sequential LLM calls (e.g., multi-hop loops), demonstrating robustness in scenarios where traditional prompt engineering struggles.

[LLM-5] Language Models are Injective and Hence Invertible

Link: <https://arxiv.org/abs/2510.15511>

Abstract: Transformer components such as non-linear activations and normalization are inherently non-injective, suggesting that different inputs could map to the same output and prevent exact recovery of the input from a model's representations. In this paper, we challenge this view. First, we prove mathematically that transformer language models mapping discrete input sequences to their corresponding sequence of continuous representations are injective and therefore lossless, a property established at initialization and preserved during training. Second, we confirm this result empirically through billions of collision tests on six state-of-the-art language models, and observe no collisions. Third, we operationalize injectivity: we introduce Siplt, the first algorithm that provably and efficiently reconstructs the exact input text from hidden activations, establishing linear-time guarantees and demonstrating exact invertibility in practice. Overall, our work establishes injectivity as a fundamental and exploitable property of language models, with direct implications for transparency, interpretability, and safe deployment.

Area 3: Diffusion

[DIFF-1] Masked Diffusion Language Models with Frequency-Informed Training

Link: arxiv.org/abs/2509.05056

A masked diffusion language modeling framework is presented for data-efficient training for the BabyLM 2025 Challenge. The approach applies diffusion training objectives to language modeling under strict data constraints, incorporating frequency-informed masking that prioritizes learning from rare tokens while maintaining theoretical validity. Multiple noise scheduling strategies, including two-mode approaches, are explored, and different noise weighting schemes within the NELBO objective are investigated. The method is evaluated on the BabyLM benchmark suite, measuring linguistic competence, world knowledge, and human-likeness. Results show performance competitive to hybrid autoregressive-masked baselines, demonstrating that diffusion-based training offers a viable alternative for data-restricted language learning.

[DIFF-2] BD3-LMs (Block Diffusion Language Models)

Link: <https://m-arriola.com/bd3lms/>

Block Discrete Denoising Diffusion Language Models (BD3-LMs) present a novel framework that interpolates between autoregressive and discrete diffusion models by decomposing token sequences into blocks and performing discrete diffusion within each block. This approach combines the strengths of both paradigms: achieving better likelihoods and flexible-length generation from autoregressive models, while maintaining fast parallel generation and controllability from diffusion models.

[DIFF-3] DUO (The Diffusion Duality)

Link: <https://s-sahoo.com/duo/>

DUO (Diffusion Duality) demonstrates that uniform-state discrete diffusion emerges from underlying Gaussian diffusion, enabling few-step generation in discrete diffusion language models by exploiting this continuous foundation. The work reveals a fundamental mathematical insight: discreteness emerges from underlying continuity, similar to how quantum mechanics produces quantized energy states from continuous wave equations. By establishing this connection, DUO enables transfer of techniques from continuous to discrete domains, including curriculum learning and consistency distillation.

Area 4: Cognitively Motivated Architectures

[COG-1] Hybrid ACNs: Unifying Auto-Compressing and Residual Architectures

Link: <https://openreview.net/pdf?id=ORSclmXC6X>

The paper proposes Hybrid Auto-Compressing Networks (H-ACNs), unifying ACNs and ResNets under a single mathematical formulation controlled by trainable scalar residual weighting parameters per layer. Through theoretical analysis, it is shown that both architectures represent points on a continuous spectrum, with traditional ACNs and ResNets as special cases. The key contribution is demonstrating that H-ACNs, when initialized close to ACNs, match ResNets training efficiency while preserving ACN-like robustness and compression capabilities. Experiments across vision transformers, MLP-mixers, and GPT-2 architectures show that H-ACNs achieve training convergence on par with ResNets, while maintaining ACNs' superior noise robustness and generalization. Furthermore, learned residual weights exhibit distinct connectivity patterns across tasks: vision tasks favor local connectivity patterns resembling early visual cortex processing, while language tasks converge to modular hierarchical inter-layer structures similar to hierarchical language processing regions. The work also examines how initialization impacts performance and connectivity, challenging the universality of the common ResNet-like initialization of residual weights. Overall, the results establish Hybrid ACNs as a practical framework for efficiently balancing training speed and representation quality, while revealing principles of how functional connectivity patterns should vary across domains, modalities, and tasks.

[COG-2] Random synaptic feedback weights support error backpropagation for deep learning

Link: <https://www.nature.com/articles/ncomms13276>

Abstract: The brain processes information through multiple layers of neurons. This deep architecture is representationally powerful, but complicates learning because it is difficult to identify the responsible neurons when a mistake is made. In machine learning, the backpropagation algorithm assigns blame by multiplying error signals with all the synaptic weights on each neuron's axon and further downstream. However, this involves a precise, symmetric backward connectivity

pattern, which is thought to be impossible in the brain. Here we demonstrate that this strong architectural constraint is not required for effective error propagation. We present a surprisingly simple mechanism that assigns blame by multiplying errors by even random synaptic weights. This mechanism can transmit teaching signals across multiple layers of neurons and performs as effectively as backpropagation on a variety of tasks. Our results help reopen questions about how the brain could use error signals and dispel long-held assumptions about algorithmic constraints on learning.

[COG-3] Spikformer V2: Join the High Accuracy Club on ImageNet with an SNN Ticket

Link: <https://arxiv.org/abs/2401.02020>

Abstract: Spiking Neural Networks (SNNs), known for their biologically plausible architecture, face the challenge of limited performance. The self-attention mechanism, which is the cornerstone of the high-performance Transformer and also a biologically inspired structure, is absent in existing SNNs. To this end, we explore the potential of leveraging both self-attention capability and biological properties of SNNs, and propose a novel Spiking Self-Attention (SSA) and Spiking Transformer (Spikformer). The SSA mechanism eliminates the need for softmax and captures the sparse visual feature employing spike-based Query, Key, and Value. This sparse computation without multiplication makes SSA efficient and energy-saving. Further, we develop a Spiking Convolutional Stem (SCS) with supplementary convolutional layers to enhance the architecture of Spikformer. The Spikformer enhanced with the SCS is referred to as Spikformer V2.

Area 5: Multimodality

[MM-1] DeepMLF: Multimodal language model with learnable tokens for deep fusion in sentiment analysis

Link: arxiv.org/abs/2504.11082

While multimodal fusion has been extensively studied in Multimodal Sentiment Analysis, the role of fusion depth and multimodal capacity allocation remains underexplored. DeepMLF is introduced as a novel multimodal language model with learnable tokens tailored toward deep fusion. DeepMLF leverages an audiovisual encoder and a pretrained decoder language model augmented with multimodal information across its layers. Learnable tokens are appended to the language model that capture modality interactions in a controlled fashion and preserve independent information flow for each modality. These fusion tokens gather linguistic information via causal self-attention in language model blocks and integrate with audiovisual information through cross-attention multimodal blocks.

[MM-2] MEDUSA: A Multimodal Deep Fusion Multi-Stage Training Framework for Speech Emotion Recognition in Naturalistic Conditions

Link: <https://arxiv.org/abs/2506.09556>

SER is a challenging task due to the subjective nature of human emotions and their uneven representation under naturalistic conditions. We propose MEDUSA, a multimodal framework with a four-stage training pipeline, which effectively handles class imbalance and emotion ambiguity. The first two stages train an ensemble of classifiers that utilize DeepSER, a novel extension of a deep cross-modal transformer fusion mechanism from pretrained self-supervised acoustic and linguistic representations. Manifold MixUp is employed for further regularization. The last two stages optimize a trainable meta-classifier that combines the ensemble predictions. Our training approach incorporates human annotation scores as soft targets, coupled with balanced data sampling and multitask learning. MEDUSA ranked 1st in Task 1: Categorical Emotion Recognition in the Interspeech 2025: Speech Emotion Recognition in Naturalistic Conditions Challenge.

Area 6: Special Topics

[SPEC-1] The Smol Training Playbook: The Secrets to Building World-Class LLMs

Link: <https://huggingface.co/spaces/HuggingFaceTB/smol-training-playbook>

Abstract: A practical journey through the challenges, decisions, and messy reality behind training state-of-the-art language models. This playbook is part of the SmolLM ecosystem from Hugging Face, providing guidance on training small language models including details about choosing baselines, training strategies, datasets (including FineWeb-Edu, Cosmopedia, and SmolLM-Corpus), and best practices for building efficient and capable small language models. The playbook covers the full training pipeline from pre-training through post-training, with specific focus on the methodologies used to create the SmolLM family of models (135M, 360M, 1.7B, and 3B parameters).

Οδηγίες για την εξαμηνιαία εργασία

→ Αναπαραγωγή των αποτελεσμάτων και πιθανές επεκτάσεις ενός ερευνητικού άρθρου στη μηχανική μάθηση μέσω πρακτικού πειραματισμού και ομαδικής συνεργασίας. Οι κύριοι **στόχοι** είναι: (1) η κριτική επαλήθευση των ισχυρισμών ενός επιλεγμένου άρθρου μέσω πειραμάτων μικρής κλίμακας χρησιμοποιώντας περιορισμένους, ελεύθερα διαθέσιμους υπολογιστικούς πόρους, (2) η διερεύνηση πιθανών επεκτάσεων ή βελτιώσεων πέραν της αρχικής εργασίας, και (3) η συνεργασία σε μια ερευνητική ομάδα μεσαίου μεγέθους.

→ Οδηγίες για τις **ενδιάμεσες παρουσιάσεις** της εξαμηνιαίας εργασίας. Η παρουσίαση θα γίνει την τελευταία εβδομάδα του εξαμήνου, θα κρατήσει **10 λεπτά** και θα συνοδεύεται από διαφάνειες (παρακαλώ να αναρτηθεί ο σύνδεσμος στις διαφάνειες στο google doc δήλωσης εξαμηνιαίας εργασίας) και θα περιέχει:

1. Σύντομη παρουσίαση / σύνοψη του paper που θα υλοποιήσετε (6-8 διαφάνειες)
2. Το πλάνο από τα βήματα που θα ακολουθήσετε (2-4 διαφάνειες)
3. Υπολογιστικοί πόροι και βάσεις δεδομένων που θα χρησιμοποιήσετε (2 διαφάνειες)
4. Με τι θα ασχοληθεί κάθε μέλος της ομάδας (πίνακας 1 διαφάνεια)

Θα ακολουθήσουν ερωτήσεις και απαντήσεις (10 λεπτά ανά παρουσίαση). Παρακαλώ να τηρηθούν τα χρονικά όρια.

→ Οδηγίες για τις **τελικές παρουσιάσεις** της εξαμηνιαίας εργασίας. Η παρουσίαση θα γίνει την τελευταία εβδομάδα της εξεταστικής, θα κρατήσει **15 λεπτά** και θα συνοδεύεται από διαφάνειες (παρακαλώ να αναρτηθεί ο σύνδεσμος στις διαφάνειες στο google doc δήλωσης εξαμηνιαίας εργασίας) και θα περιέχει:

1. Πολύ σύντομη παρουσίαση / σύνοψη του paper (1-2 διαφάνειες)
2. Τα αποτελέσματά σας (8-10 διαφάνειες)
3. Αποτίμηση της δουλειάς σας και μελλοντικές επεκτάσεις (1-2 διαφάνειες)
4. Με τι θα ασχοληθεί κάθε μέλος της ομάδας (πίνακας 1 διαφάνεια)
5. Retrospective: Τι μας δυσκόλεψε και θα μπορούσαμε να είχαμε κάνει καλύτερα (1 διαφάνεια)

Θα ακολουθήσουν ερωτήσεις και απαντήσεις (5 λεπτά ανά παρουσίαση). Παρακαλώ να τηρηθούν τα χρονικά όρια.

→ Οδηγίες για **τελική αναφορά**: Η τελική αναφορά θα είναι σύντομη και περιεκτική στην μορφή paper 4-5 δίστηλων σελίδων ή 6-8 μονόστηλων σελίδων. Θα έχει την κλασσική μορφή ερευνητικού paper με έμφαση στην καινοτομία της μεθόδου σας και αναλυτική περιγραφή των αποτελεσμάτων σας.

→ **Βαθμολόγηση**:

1. Τουλάχιστον 2 συναντήσεις με τους επιστημονικούς συνεργάτες 10%
2. Ενδιάμεση παρουσίαση 20%
3. Τελική παρουσίαση 40%
4. Τελική αναφορά 30%

Οι εργασίας που ξεχωρίζουν θα πάρουν bonus 10-20%. Κύρια κριτήρια για το bonus είναι η καινοτομία, ερευνητικά αποτελέσματα πέραν αυτών που αναφέρονται στο paper που υλοποιήσατε και η καλή συνεργασία ανάμεσα στα μέλη της ομάδας.

→ **Σημαντικά θέματα:**

1. Η **συνεργασία** ανάμεσα στα μέλη της ομάδας σας δεν θα είναι πάντα εύκολη λόγω του μεγέθους της ομάδας σας, διαφορετικών ωραρίων και υποχρεώσεων. Πέραν των συναντήσεων, είναι σημαντικό να έχετε τακτική επικοινωνία, π.χ., μέσω slack ή αντίστοιχου εργαλείου, όπου θα περιγράφετε την καθημερινή σας πρόοδο σε όλα τα μέλη της ομάδας.
2. Πρέπει να είσαστε πολύ προσεκτικοί ως προς το μέγεθος των μοντέλων και των δεδομένων που θα χρησιμοποιήσετε γιατί δεν θα σας δοθούν **υπολογιστικοί πόροι**. Οπότε από την αρχή πρέπει να είστε σίγουροι ότι μπορείτε να τρέξετε τα πειράματά σας με δωρεάν πόρους όπως το google colab <https://colab.research.google.com/>

Instructions in English

→ **Goal:** Reproduction of results and possible extensions of a research paper in machine learning through hands-on experimentation and team collaboration. The main objectives are: (1) critical verification of the claims of a selected paper through small-scale experiments using limited, freely available computational resources, (2) exploration of potential extensions or improvements beyond the original work, and (3) collaboration in a mid-sized research team.

→ **Guidelines for the interim presentations of the semester project:** The presentation will take place during the last week of the semester, will last 10 minutes, and must be accompanied by slides (please upload the link to your slides in the Google Doc where you declared your semester project). It should include:

1. A brief presentation/summary of the paper you will implement (6–8 slides)
2. A plan outlining the steps you will follow (2–4 slides)
3. Computational resources and datasets you will use (2 slides)
4. A breakdown of what each team member will be working on (table, 1 slide)

A Q&A session will follow (10 minutes per presentation). Please respect the time limits.

→ **Guidelines for the final presentations of the semester project:** The final presentation will take place during the last week of the exam period, will last 15 minutes, and must be accompanied by slides (please upload the link to your slides in the Google Doc where you declared your semester project). It should include:

1. A very brief presentation/summary of the paper (1–2 slides)
2. Your method and results (8–10 slides)
3. Evaluation of your work and potential future extensions (1–2 slides)
4. A breakdown of what each team member worked on (table, 1 slide)
5. Retrospective: What challenged us and what we could have done better (1 slide)

A Q&A session will follow (5 minutes per presentation). Please respect the time limits.

→ **Final Report Guidelines:** The final report should be concise and in the format of a research paper: 4–5 pages (two-column layout) or 6–8 pages (single-column layout). It should follow the typical structure of a research paper, with an emphasis on the innovation of your method and a detailed description of your results.

→ **Grading:**

1. At least 2 meetings with academic supervisors: 10%
2. Interim presentation: 20%
3. Final presentation: 40%
4. Final report: 30%

Outstanding projects will receive a bonus of 10–20%. The main criteria for the bonus are: innovation, research results that go beyond what is presented in the implemented paper, and good collaboration among team members.

→ **Important Notes:**

- Collaboration among team members may not always be easy due to the size of the team, varying schedules, and other obligations. Beyond scheduled meetings, it's important to maintain regular communication—e.g., via Slack or a similar tool—where everyone can share their daily progress with the whole team.
- You must be very careful about the size of the models and datasets you use, as no computational resources will be provided. Therefore, from the start, ensure you can run your experiments using free resources such as [Google Colab](#).